Multiscale Heterogeneity of an Urban PM2.5 Monitoring Network: Wavelet-Based Functional Inference for Lima, Peru

Heterogeneidad multiescala de una red urbana de monitoreo de PM2.5: inferencia funcional wavelet para Lima, Perú

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Abstract

Urban PM2.5 networks are often evaluated through mean concentrations or exceedance counts, which cannot reveal when stations provide distinct information. We analyse hourly PM2.5 at five REMCA stations in Lima, Peru (1 January 2022–28 February 2023; $n = 10{,}176$) with a wavelet framework combining conservative preprocessing and assumption-checked inference. Missing values were interpolated (2.4–6.8%). Isolated instrumental errors were replaced only when two residual rules flagged them and no other station corroborated them (69–110 h per station; 0.68–1.08%), compared with 8–13% under STL+3×IQR. A Symmlet-7 transform on mirror-padded series (2^{14}) supported tests on central coefficients at $j = 1, \ldots, 9$ (Bonferroni $M = 9$) using repeated-measures ANOVA, Friedman, or block-permutation tests ($N_{\text{perm}} = 9{,}999$). Differences are strong at $j = 1$–7, peak in the diurnal band $j = 4$ (Kendall $W = 0.69$), and lose Bonferroni support at $j = 8$; no evidence of difference at $j = 9$, and $j \geq 10$ was not formally testable. Diurnal energy is 20.4% at San Juan de Lurigancho versus 12.4% at San Borja. Sensitivity preserves the $j = 1$–7 pattern. The results support denser short-scale coverage in high-contrast corridors and caution against treating multi-week signals as station-specific.

Keywords: wavelet transform; PM2.5; multiscale analysis; functional ANOVA; air quality monitoring.

Resumen. Las redes de PM2.5 suelen evaluarse mediante concentraciones medias o recuentos de excedencias, que no revelan las escalas temporales en las que las estaciones aportan información distinta. Analizamos PM2.5 horario en cinco estaciones REMCA de Lima (1 enero 2022–28 febrero 2023; $n = 10{,}176$) con un marco wavelet multiescala que combina preprocesamiento conservador e inferencia condicionada a diagnósticos. Se imputaron faltantes (2,4–6,8 %) y solo se sustituyeron errores instrumentales aislados cuando dos reglas de residuos los señalaron y ninguna otra estación los corroboró (69–110 h; 0,68–1,08 %), frente al 8–13 % de STL+3×IQR. Con DWT Symmlet-7 y pruebas en $j = 1, \ldots, 9$ (Bonferroni $M = 9$; $N_{\text{perm}} = 9{,}999$), las diferencias son fuertes en $j = 1$–7, máximas en la banda diurna $j = 4$ ($W = 0{,}69$) y pierden soporte Bonferroni en $j = 8$; no hay evidencia de diferencia en $j = 9$ y $j \geq 10$ no fue testeable formalmente. La energía diurna es 20,4 % en SJL y 12,4 % en SBJ. La sensibilidad confirma el patrón. Los resultados apoyan mayor cobertura en escalas cortas de corredores contrastados y cautela al interpretar señales multi-semana como específicas de estación.

Palabras clave: transformada wavelet; PM2.5; análisis multiescala; ANOVA funcional; monitoreo de calidad del aire.

1. Introduction

Fine particulate matter (PM2.5) remains a major urban public-health burden. In Peru, ambient PM2.5 is associated with approximately 230 premature deaths per million inhabitants, among the highest rates in the Americas [14]. Metropolitan Lima has long ranked among the most polluted large cities in Latin America for particulate matter [8], with multi-station analyses documenting persistent exceedances of national and WHO guidelines and strong spatial contrasts across northern, eastern and southern districts [15–18]. Lima faces recurrent wintertime deterioration driven by traffic and industrial emissions and by meteorological regimes that suppress vertical mixing [15, 16].

SENAMHI's REMCA network records hourly PM2.5 at stations distributed across the metropolitan area. Exposure-mapping and network studies in Lima show that sparse monitors and strong intra-urban gradients make station placement and redundancy non-trivial design questions [17, 22]. Operational analyses often reduce these series to daily means, exceedance counts, or classical ANOVA on scalars. Such summaries are sensitive to outliers and right skew, and—more importantly—they cannot locate the *timescales* of spatial heterogeneity [12, 20]. Functional data analysis treats each station as a curve rather than a single number [5, 7]. Wavelet bases further localise differences jointly in scale and time [1, 4], and have been used to characterise monitoring-site similarity and local-versus-background variability in air-quality networks [19, 21]. Wavelet-based functional ANOVA (wavelet-FANOVA) projects curves onto detail coefficients so that scale-wise tests can be stated in finite dimension [2, 3, 9]. What remains missing for Lima is a scale-resolved assessment of whether stations are redundant for specific monitoring objectives, rather than only whether their marginal distributions differ.

Two practical difficulties arise with environmental series. First, aggressive residual fences (e.g. STL+3×IQR) can replace a large share of high concentrations that may be real pollution

episodes; because multiscale energy uses squared coefficients, such replacements can distort the scales under study. Second, treating all wavelet coefficients as independent replicates ignores temporal dependence within scales and can produce overconfident p-values. Both problems can change the scale at which differences appear, and both must be addressed before inference is used for network design.

This paper addresses both issues for Lima REMCA data. We make three contributions. First, we apply a conservative instrumental-error protocol that preserves multi-hour and multi-station excursions while documenting every replacement. Second, we restrict formal inference to central coefficients at scales with adequate information ($j = 1, \ldots, 9$) and select repeated-measures ANOVA, Friedman, or block-permutation tests according to Ljung–Box and residual-normality diagnostics. Third, we translate the scale-wise evidence into an explicit decision matrix that links timescales to monitoring objectives. The analysis is reproducible from open REMCA data and a documented Python pipeline. Section 2 describes the data and methods, Section 3 reports the multiscale evidence and its implications, and Section 4 concludes.

2. Experimentation

2.1. Study area and data

Hourly PM2.5 concentrations ($\mu g\,m^{-3}$) were obtained for five REMCA stations: San Borja (SBJ), Santa Anita (STA), Villa María del Triunfo (VMT), San Juan de Lurigancho (SJL), and San Martín de Porres (SMP), from 1 January 2022 to 28 February 2023 (10,178 raw rows). SBJ and SMP represent moderate-traffic residential settings; STA and SJL high-density corridors; and VMT a peri-urban mix of sources.

2.2. Data preprocessing

The pipeline deliberately separates missing-value completion from conservative instrumental-error treatment (Fig. 1).

Alignment. Duplicate timestamps were averaged by the mean of available station values. Series were reindexed to a complete hourly grid of $n_0 = 10{,}176$ hours spanning the study window.

Imputation only (Base A). Missing values were filled by time interpolation: SBJ 693, STA 570, VMT 404, SJL 500, and SMP 244 hours (2.4–6.8%). No high-value substitution was applied at this stage.

Conservative cleaning (Base B; primary analysis). Robust STL with period 24 h was fit to each station [11]. Residuals were screened with (i) a medcouple-adjusted boxplot ($k = 1.5$) and (ii) a centred Hampel filter (window 49 h, threshold 4 MAD). An hour entered the final candidate set only if both rules flagged it, the dual-flag run length was at most one hour, and no other station showed a same-direction Hampel excursion ($K_{\text{conc}} = 2.5$) at that time. Candidates were replaced by STL trend+seasonal components and clipped at zero. Final replacements were SBJ 69, STA 81, VMT 88, SJL 72, and SMP 110 hours (0.68–1.08%). By contrast, applying STL+3×IQR on the same residuals would replace 793–1,342 points (8–13%).

Sensitivity. All formal tests were repeated on Base A and on an aggressive Base C (STL+3×IQR). Agreement of Bonferroni decisions between A and B was complete across $j = 1, \ldots, 9$; Base C differed only at $j = 8$, which became significant solely under aggressive cleaning.

Padding. Each series was mirror-reflected to length $n = 2^{14} = 16{,}384$ with $b = 3{,}104$ points per side, as required by a fully dyadic discrete wavelet transform (DWT).

2.3. Discrete wavelet transform

Padded series were decomposed with the Symmlet-7 (least-asymmetric Daubechies) wavelet in periodisation mode to $J_{\text{desc}} = 13$ detail levels for description [6, 10, 13]. Numerical quality was verified by inverse reconstruction and Parseval balance: maximum reconstruction errors were of order 10^{-10} and relative energy discrepancies of order 10^{-13} at all stations. These checks confirm implementation correctness; they do not validate distributional assumptions.

Detail energy at scale j is

$$E_j = \sum_k d_{j,k}^2, \qquad j = 1, \ldots, 13, \tag{1}$$

where $d_{j,k}$ are detail coefficients. With hourly sampling, scale j spans periods 2^j–2^{j+1} hours; thus $j = 4$ is the diurnal band (16–32 h). For this length and filter, the maximum useful DWT level is 10; levels $j \geq 11$ are retained only descriptively.

For inference, coefficients whose nominal centres fall outside the observed time window (pure padding) were discarded. Formal testing further required $j \leq 10$ and at least 12 central coefficients, which restricts inference to $j = 1, \ldots, 9$.

2.4. Assumption-checked scale-wise tests

At each formal scale, central squared coefficients were aggregated into non-overlapping temporal blocks (sizes $1, 2, 4, \ldots$) until Ljung–Box tests on $\log(1 + \cdot)$ block means, Holm-adjusted across the five stations, did not reject independence, or the largest feasible aggregation was reached. The selected aggregation corresponded to approximately 512 h blocks at $j = 1$–7, 256 h at $j = 8$, and 512 h at $j = 9$. Residual normality for a paired additive model (block + station) was assessed via Shapiro–Wilk (subsample capped at 5,000 residuals), skewness, and excess kurtosis; normality was accepted when the Shapiro–Wilk p-value was at least 0.05, $|\text{skewness}| < 1$, and $|\text{excess kurtosis}| < 2$.

Test choice followed the diagnostics:

- independence and normality acceptable → repeated-measures ANOVA (Mauchly sphericity; Greenhouse–Geisser correction if needed; generalised η^2);
- independence acceptable, normality not → Friedman test (Kendall W);
- residual dependence → block-permutation Friedman of station labels within temporal super-blocks ($N_{\text{perm}} = 9{,}999$; Kendall W).

Family-wise error was controlled by Bonferroni over $M = 9$ tests ($\alpha/9 \approx 0.00556$); Holm p-values are reported in the supplementary numerical tables. Classical coefficient-wise Kruskal–Wallis and block-MANOVA without dependence control are *not* used as primary claims: with one long curve per station, blocking alone does not create independent multivariate replicates [2, 9].

Table 1: Descriptive statistics of hourly PM2.5 ($\mu g\,m^{-3}$) after conservative preprocessing ($n =$ 10,176 per station).

	SBJ	STA	VMT	SJL	SMP
Mean	24.4	36.5	34.4	38.6	22.2
SD	14.0	21.2	18.4	20.3	11.5
CV (%)	57.4	58.0	53.6	52.5	51.6
Min	3.8	4.3	2.3	5.6	5.0
P50	20.6	31.7	30.8	34.5	19.3
P99	79.9	108.4	93.4	102.4	61.2
Max	117.1	191.5	164.9	166.4	106.1
NAs imputed	693	570	404	500	244
Outliers replaced	69	81	88	72	110

2.5. Software and reproducibility

Analyses were implemented in Python 3 (pandas, NumPy, statsmodels, PyWavelets, SciPy, pingouin) with fixed seed 2026. A reproducibility package containing the processed bases, the row-level audit, the configuration file, the analysis scripts, the figure sources and automated checks of the reported numbers accompanies this manuscript and is available from the corresponding author. Primary results use Base B. Raw REMCA data are public.

3. Results and Discussion

3.1. Descriptive characterisation

Table 1 summarises Base B. Station means range from 22.2 $\mu g\,m^{-3}$ (SMP) to 38.6 $\mu g\,m^{-3}$ (SJL), with coefficients of variation 51.6–58.0% reflecting strong right skew. Maxima remain high (e.g. STA 191.5 $\mu g\,m^{-3}$) because genuine pollution peaks were largely retained.

Figure 1A shows the shared winter enhancement of mid-2022 and persistently higher levels at SJL/STA relative to SBJ/SMP. Panel B contrasts missing-value imputation (hundreds of hours) with the sparse conservative replacements (tens of hours), documenting how little of the series is altered beyond gap filling—about an order of magnitude fewer substitutions than under STL+3$\times$IQR.

The city-mean diurnal cycle (Fig. 2A) peaks at 09:00 (41.7 $\mu g\,m^{-3}$) and troughs at 17:00 (23.8 $\mu g\,m^{-3}$). At 09:00, SJL and STA reach 54.9 and 51.6 $\mu g\,m^{-3}$, respectively, whereas SBJ and SMP remain near 29.7 $\mu g\,m^{-3}$. Panel B emphasises right skew relative to the WHO 24 h guideline of 50 $\mu g\,m^{-3}$ (shown only as a reference line on hourly distributions, not as an hourly standard): P99 values are 108.4 (STA) and 102.4 (SJL) $\mu g\,m^{-3}$, versus 61.2 at SMP. Pairwise Pearson correlations range from 0.54 (VMT–SMP) to 0.86 (STA–SJL), indicating shared regional forcing modulated by local factors.

3.2. Multiscale energy profiles

Figure 3A displays $\log_{10}(E_j)$ for $j = 1,\ldots,13$. Fine scales ($j =$ 1–4) separate high-energy stations (SJL, STA) from lower-energy sites (SBJ, SMP). In the diurnal band $j = 4$, the share of total detail energy is 20.4% (SJL), 18.7% (STA), 17.4% (SMP), 13.3% (VMT), and 12.4% (SBJ). Intermediate scales still show separation that shrinks toward longer periods. Grey shading marks levels above the

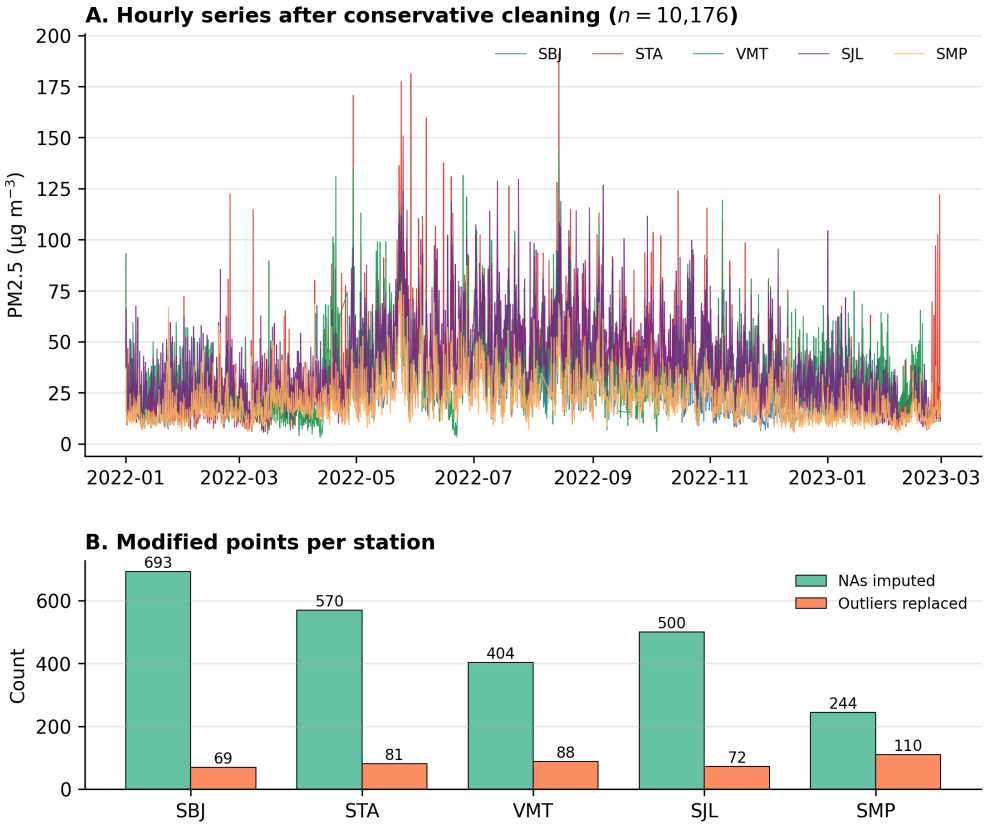


Figure 1: Conservative preprocessing of hourly PM2.5 at five Lima REMCA stations. **(A)** Aligned series ($n = 10{,}176$) after imputation and sparse instrumental-error replacement (winter peak June–August 2022; SJL/STA above SBJ/SMP). **(B)** Counts of time-interpolated missing values (teal) and conservative replacements (coral). Replacements remain ∼0.7–1.1% of each series.

useful DWT maximum; those coarse scales are exploratory only.

3.3. Scale-wise hypothesis tests

Table 2 and Fig. 3B summarise assumption-checked tests. Block permutation was required at $j = 1$–4, 6, and 7 because dependence persisted after aggregation; repeated-measures ANOVA applied at $j = 5$ (independence and normality acceptable); Friedman tests at $j = 8$–9 (independence acceptable, normality not).

Bonferroni-significant differences appear at $j = 1$–7 (about 2 h to ∼11 days). For block-permutation scales, the smallest attainable p-value with $N_{\text{perm}} = 9{,}999$ is $1/(9{,}999+1) = 1.0 \times 10^{-4}$; the table reports this value explicitly rather than as an open inequality. The largest rank-based effect is at $j = 4$ ($W = 0.69$; 95% block-bootstrap CI 0.60–0.80). At $j = 8$ the Friedman $p = 0.0079$ does not survive Bonferroni; at $j = 9$ there is no evidence of difference ($p = 0.185$). We do not interpret $j \geq 10$ as equality of stations: those scales lack enough central coefficients for reliable formal tests and are influenced by boundary padding. Secondary pairwise Wilcoxon tests (Holm within scale) at $j = 1$–7 separated the high-contrast corridor stations from the residential sites in most pairs; at $j = 4$, eight of ten pairs differed, the exceptions being SJL–STA and SBJ–SMP. These tests are descriptive and do not replace the omnibus Bonferroni decision; pairwise results at $j = 8$–9 are reported for

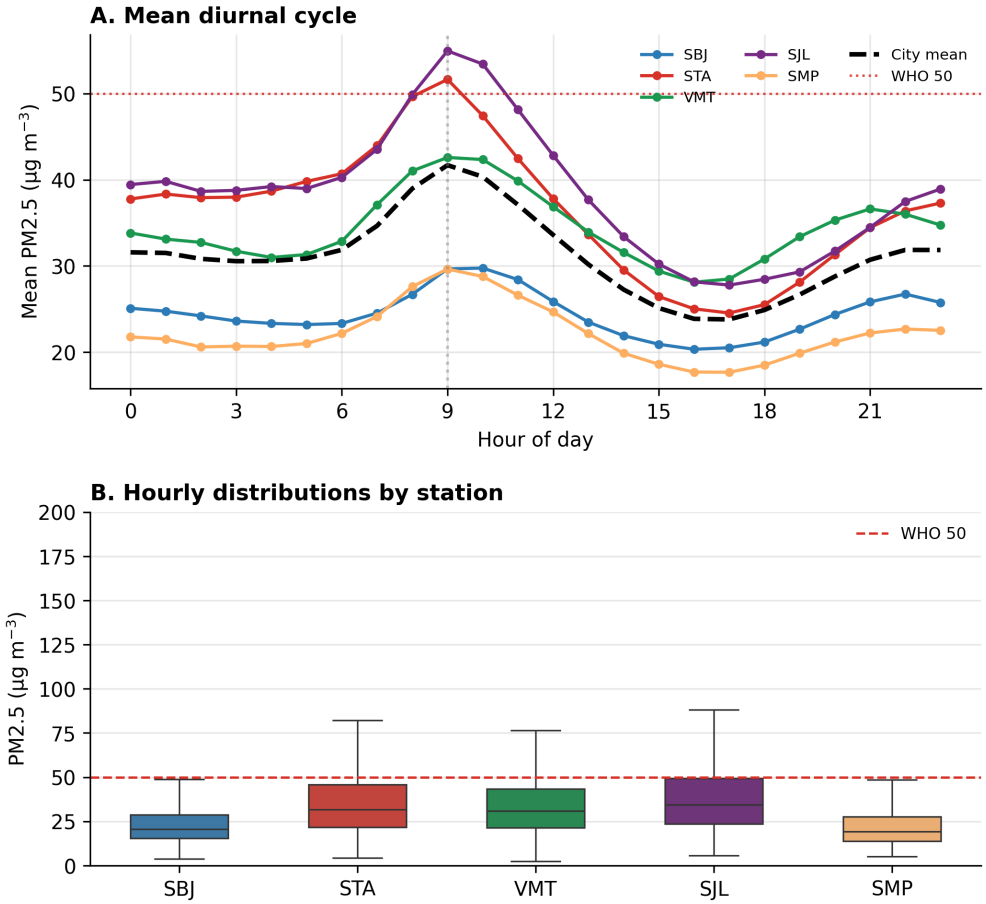


Figure 2: Diurnal cycle and distributions on Base B. **(A)** Mean hourly concentration by station and city mean (dashed); morning peak at 09:00, afternoon trough at 17:00. **(B)** Hourly distributions; dashed red line: WHO 50 μg m^{-3} guideline.

completeness but not interpreted.

Relative to coefficient-wise Kruskal–Wallis analyses that report extreme p-values under independence assumptions, the present p-values are more conservative, but the scientific ordering—strong short-scale heterogeneity, weaker intermediate scales, no formal evidence at the longest tested band—is retained. Temporal aggregation produced 19–20 blocks at $j = 1$–7, 40 at $j = 8$, and 20 at $j = 9$. These counts are adequate for the omnibus tests but limited for long-scale equivalence claims; the non-significant $j = 9$ result is therefore reported as absence of evidence, not as evidence of absence. Sensitivity: Bases A and B share identical Bonferroni decisions; Base C (aggressive) additionally flags $j = 8$, illustrating how heavy peak suppression can alter intermediate-scale inference.

3.4. Diurnal band energy allocation

The diurnal band concentrates the strongest rank effect and a clear spatial gradient of energy share (SJL 20.4%, 95% CI 17.4–22.9, versus SBJ 12.4%, 95% CI 10.3–16.5; the intervals do not overlap), consistent with larger morning peaks at SJL/STA (Fig. 2A) and denser traffic corridors. Aggressive cleaning reduces diurnal energy especially at SBJ (to 7.8%), confirming that peak suppression can shrink the very contrast of engineering interest.

Table 2: Scale-wise inter-station tests after assumption checking (Base B). Bonferroni threshold $p < \alpha/9 \approx 0.0056$. Effect: Kendall W or generalised η^2_G. For block-permutation scales, $p = 1.0 \times 10^{-4}$ is the minimum attainable value, $1/(N_{\text{perm}} + 1)$.

j	Period (h)	Method	p	p_{Bonf}	Sig.	Effect
1	2–4	Block perm.	1.0×10^{-4}	9.0×10^{-4}	Yes	W=0.61
2	4–8	Block perm.	1.0×10^{-4}	9.0×10^{-4}	Yes	W=0.62
3	8–16	Block perm.	1.0×10^{-4}	9.0×10^{-4}	Yes	W=0.65
4	16–32	Block perm.	1.0×10^{-4}	9.0×10^{-4}	Yes	W=0.69
5	32–64	ANOVA-RM	1.4×10^{-11}	1.3×10^{-10}	Yes	η^2_G=0.21
6	64–128	Block perm.	1.0×10^{-4}	9.0×10^{-4}	Yes	W=0.35
7	128–256	Block perm.	1.0×10^{-4}	9.0×10^{-4}	Yes	W=0.45
8	256–512	Friedman	7.9×10^{-3}	7.1×10^{-2}	No	W=0.09
9	512–1024	Friedman	0.185	1.000	No	W=0.08

3.5. Interpretation of scale-specific heterogeneity

Scalar ANOVA on hourly values can establish mean differences but not their timescale. Assumption-checked wavelet tests show that heterogeneity is concentrated from a few hours to about ten days, is strongest in the diurnal band, and loses formal support by $j = 9$ ($\sim$21–43 days). This matches a physical partition: local emissions and boundary-layer cycles dominate short scales, whereas basin-wide meteorology increasingly synchronises stations at longer scales. We deliberately avoid claiming that stations are "indistinguishable" at seasonal resolutions: those scales were not formally tested with adequate central coefficients.

3.6. Practical implications for monitoring networks

The results support a simple engineering decision rule. For short-term exposure and diurnal compliance diagnostics, spatial density should be preserved or increased in high-contrast corridors (SJL/STA): morning peaks reach $\approx 55\ \mu\text{g m}^{-3}$ versus $\approx 30\ \mu\text{g m}^{-3}$ at SBJ/SMP, and diurnal energy differs by about eight percentage points. For multi-week or basin-scale tracking, additional stations add less independent information once $j \geq 8$, so a smaller representative subset may suffice for those objectives. In operational terms: prioritise dense, reliable sampling where $j = 1$–4 contrasts are large; avoid interpreting weak multi-week differences as evidence that stations are interchangeable for acute management. Table 3 summarises these implications by monitoring objective.

3.7. Methodological considerations

Conservative preprocessing addresses right-skewed PM2.5 tails: only 0.7–1.1% of hours are replaced, and sensitivity shows complete Bonferroni agreement between imputation-only and conservative bases. Adaptive tests address dependence that classical Kruskal–Wallis on all $d^2_{j,k}$ ignores. Limitations remain. The record covers five stations and 14 months, so the decision matrix is a case-specific engineering input rather than a universal design rule. One long curve per station limits the number of independent replicates; Hampel and medcouple thresholds were fixed a priori; and coarse scales are affected by boundary padding. No meteorological covariates or source apportionment are used, so the physical interpretation identifies plausible mechanisms rather than attributing sources. The former aggressive 8–13% replacement rule is retained only as a contrast arm, not as the primary scientific analysis.

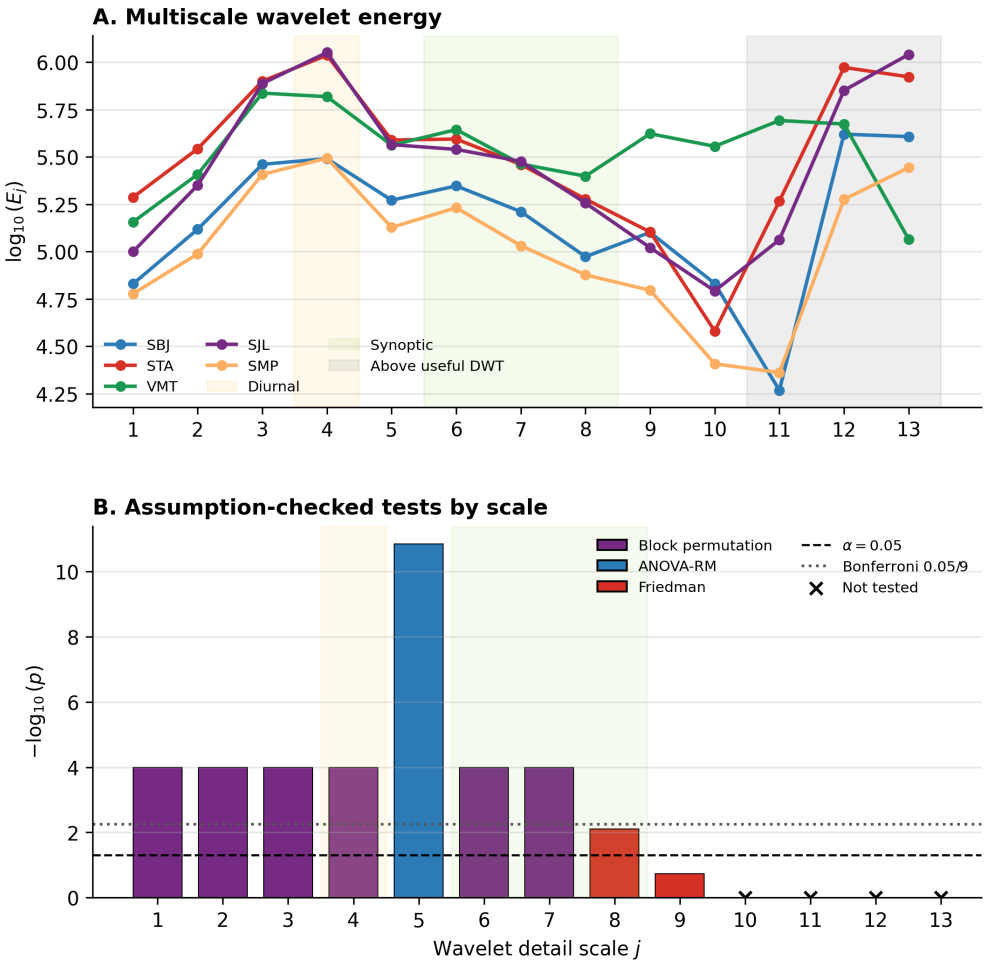


Figure 3: Multiscale energy and assumption-checked inference (Base B). **(A)** Per-station $\log_{10}(E_j)$ with diurnal (gold) and synoptic (green) bands; grey: above useful DWT level. **(B)** $-\log_{10}(p)$ by selected method (block permutation, ANOVA-RM, or Friedman). Dashed: $\alpha = 0.05$; dotted: Bonferroni $0.05/9$; crosses: scales not tested ($j \geq 10$).

Table 3: Evidence-based decision matrix for monitoring effort. Timescales follow the DWT bands of Table 2; recommendations are conditional on the five-station, 14-month evidence base.

Monitoring objective	Relevant scales	Evidence	Implication
Acute and diurnal exposure	$j = 1$–4 (2–32 h)	Strongest inter-station differences; W up to 0.69; 09:00 peaks ≈55 versus ≈30 μg m^{-3} (SJL/STA versus SBJ/SMP)	Preserve hourly resolution and prioritise short-scale coverage in high-contrast corridors
Synoptic and weekly tracking	$j = 5$–7 (32 h–11 d)	Bonferroni-significant but weaker effects ($W = 0.35$–0.45; $\eta^2_G = 0.21$)	Retain spatial coverage; interpret station-specific weekly anomalies cautiously
Multi-week and seasonal	$j = 8$–9 (10–43 d)	No Bonferroni evidence; $j = 9$ has low power (20 blocks)	Do not infer station redundancy; use dedicated seasonal or trend analyses
Long-term trends	$j \geq 10$	Not formally testable with this series length	Use a separate trend model; not a basis for network reduction

3.8. Comparison with previous work

Functional and multiscale analyses of monitoring networks have been used in Europe and elsewhere to identify representative sites and redesign sparse networks [19, 20], while wavelet decompositions help separate local events from persistent enhancements in urban sensor systems [21]. Wavelet-FANOVA theory [2, 9] has had limited operational use on Latin American air-quality networks. The combination of assumption-checked wavelet inference with an explicit scale-to-objective decision matrix has, to our knowledge, not been reported for a Latin American PM2.5 network. The present study supplies a reproducible pipeline for Lima that ties residual diagnostics to test choice and documents how preprocessing changes multiscale conclusions relevant to network design [15, 16].

4. Conclusions

For Lima REMCA hourly PM2.5 (Base B), inter-station differences are Bonferroni-significant at scales $j = 1$–7, strongest at the diurnal band ($j = 4$, Kendall $W = 0.69$), not Bonferroni-significant at $j = 8$, and non-significant at $j = 9$; $j \geq 10$ cannot be tested formally with this record length. Diurnal energy fractions range from 12.4% (SBJ) to 20.4% (SJL). Conservative cleaning changes fewer than about 1.1% of hours; sensitivity to imputation-only data leaves the main significance pattern unchanged, whereas aggressive STL+3$\times$IQR can alter intermediate-scale calls. The resulting decision matrix supports denser short-scale monitoring in high-contrast corridors and cautions against inferring multi-week redundancy. These conclusions are conditional on five stations and a 14-month record; extending the framework to other networks should revisit the preprocessing and block structure.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors' Contributions

H.S. Aquino-Mosquera: Conceptualization, Methodology, Software, Formal analysis, Writing—original draft. J.E. Villanueva-Ramírez: Data curation, Validation, Writing—review & editing. D.A. Huamán-Cárdenas: Statistical analysis, Software, Visualization. G.R. Quispe-Mendoza: Supervision, Investigation, Writing—review & editing.

Data Availability Statement

Hourly PM2.5 data are publicly available from SENAMHI REMCA (https://www.senamhi.gob.pe). Processed bases, the row-level preprocessing audit, analysis scripts, configuration files and figure sources are available from the corresponding author and at https://github.com/jelincovil/pm25-lima-wavelet; the release will be archived with a DOI upon acceptance. Correspondence: Henry S. Aquino-Mosquera (henry.aquino@ciencia.org.pe).

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